HOSPITAL BILLING AND PATIENT CLASSIFICATION

Python Project Report

1.Overview

Hospital Billing and Patient Classification is a system designed to calculate hospital bills and classify patients based on their treatment type and services used. The system collects patient details, consultation fees, laboratory charges, medicine costs, room charges, and other medical expenses. It calculates the total bill and categorizes patients according to predefined hospital policies. This system reduces manual calculations, minimizes billing errors, saves time, and improves hospital management.

2.Example Policy Used

The following policies are used in the Hospital Billing and Patient Classification system:

1. Every patient must have a valid patient ID.
2. Patient details and treatment information must be entered correctly.
3. Patients are classified into Outpatient (OP), Inpatient (IP), or Emergency categories.
4. Consultation fees are charged according to the patient category.
5. Laboratory tests, medicines, and room charges are added to the bill when applicable.
6. A room charge is applied only to patients who use hospital rooms.
7. The total bill is calculated by adding all applicable charges.
8. Any discount is applied according to the hospital's predefined policy.
9. The final bill amount must not be negative.
10. The system displays the patient category, individual charges, discount, and final bill amount.

Note: The charges and policies used in this project are sample values for academic purposes.

3.Problem Statements

Manual hospital billing requires considerable time and effort to calculate consultation fees, medicine costs, laboratory charges, and room charges. Manual calculations may lead to errors in the final bill amount. Classifying patients based on their treatment category can also become difficult when patient records are maintained manually.

Therefore, a computerized system is needed to classify patients, calculate hospital charges accurately, apply predefined discounts, and generate the final bill efficiently.

4.Objectives

The main objectives of the Hospital Billing and Patient Classification system are:

1. To maintain basic patient information.
2. To classify patients as Outpatient, Inpatient, or Emergency.
3. To calculate consultation fees automatically.
4. To calculate laboratory, medicine, and room charges.
5. To calculate the total hospital bill accurately.
6. To apply discounts according to predefined policies.
7. To reduce manual calculation errors.
8. To save time in hospital billing operations.
9. To display a clear and detailed bill for each patient.
10. To improve the efficiency of hospital administration.

5.Assumptions

The following assumptions are considered in this project:

1. Each patient has a unique patient ID.
2. Patient details are entered correctly.
3. Each patient belongs to a valid patient category.
4. Consultation fees are predefined for each category.
5. Laboratory and medicine charges are entered as valid amounts.
6. Room charges are applicable only to patients who use hospital rooms.
7. Discounts are applied according to predefined rules.
8. All charges are non-negative values.
9. The system uses the same billing rules for all patients in the same category unless a specific policy applies.
10. The program calculates bills based on the entered information and sample hospital policies.

6.Algorithm

Step 1: Start.

Step 2: Enter the patient ID and patient details.

Step 3: Enter the treatment category.

Step 4: Check whether the patient ID is valid.

Step 5: If the patient ID is invalid, display an error message.

Step 6: If the category is OP, classify the patient as Outpatient.

Step 7: If the category is IP, classify the patient as Inpatient.

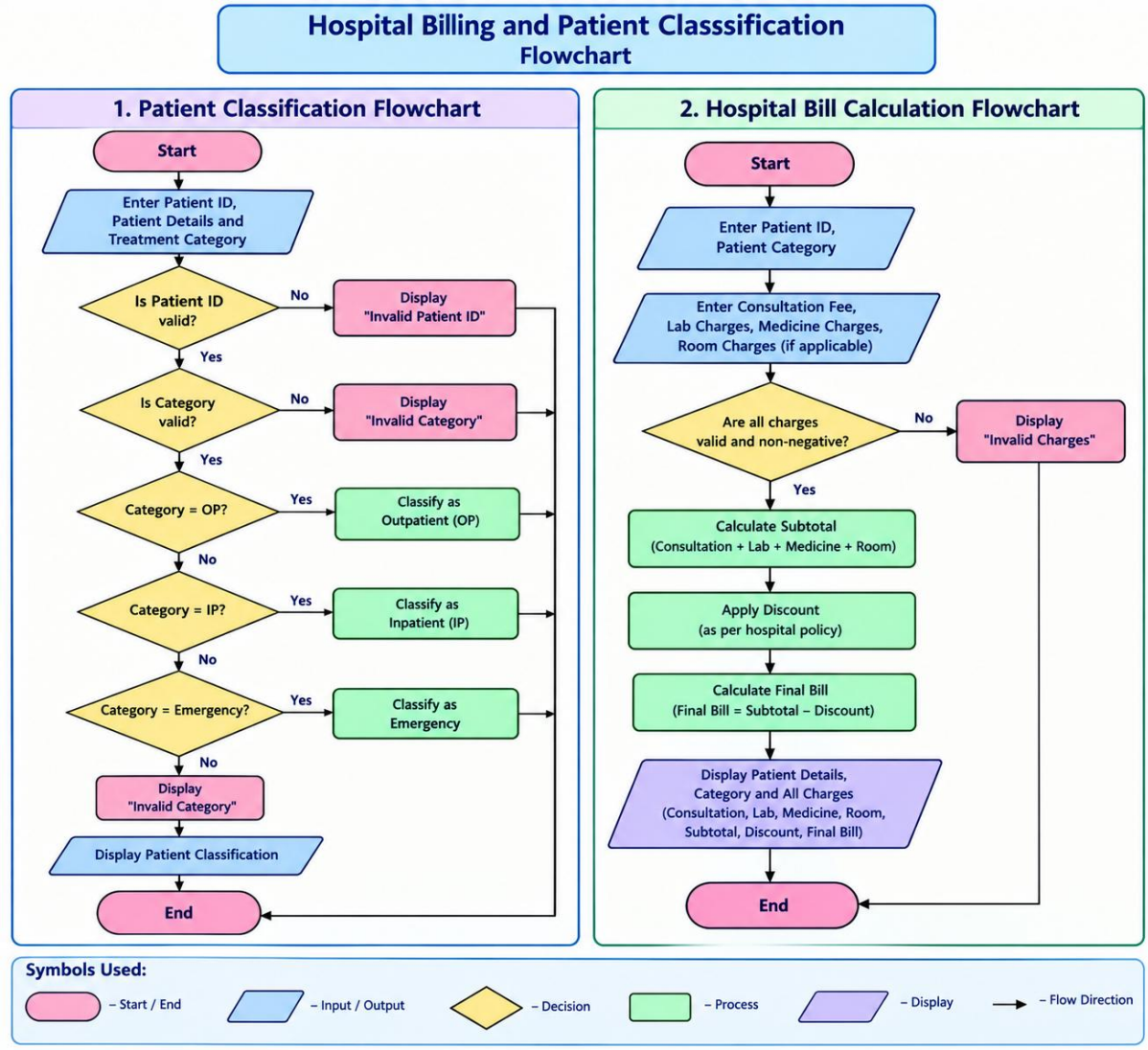
Step 8: If the category is Emergency, classify the patient as an Emergency patient.

Step 9: If the category is invalid, display an invalid category message.

Step 10: Display the patient classification.

Step 11: Stop.

7.Flowchart



8.Pseudocode

Pseudocode for Patient Classification

BEGIN

INPUT Patient_ID

INPUT Patient_Name

INPUT Patient_Category

IF Patient_ID is invalid THEN

DISPLAY "Invalid Patient ID"

```
  ELSE IF Patient_Category = "OP" THEN
    DISPLAY "Patient Classified as Outpatient"

  ELSE IF Patient_Category = "IP" THEN
    DISPLAY "Patient Classified as Inpatient"

  ELSE IF Patient_Category = "Emergency" THEN
    DISPLAY "Patient Classified as Emergency"

  ELSE
    DISPLAY "Invalid Patient Category"

  END IF

END
```

Pseudocode For Hospital Bill Calculation

```
BEGIN

  INPUT Patient_ID
  INPUT Patient_Category
  INPUT Consultation_Fee
  INPUT Lab_Charges
  INPUT Medicine_Charges
  INPUT Room_Charges

  IF Any charge is negative THEN
    DISPLAY "Invalid Charges"
```

```
    ELSE
        Subtotal = Consultation_Fee
                + Lab_Charges
                + Medicine_Charges
                + Room_Charges

        INPUT Discount_Percentage

        Discount = Subtotal * Discount_Percentage / 100

        Final_Bill = Subtotal - Discount

        DISPLAY Patient_ID
        DISPLAY Patient_Category
        DISPLAY Consultation_Fee
        DISPLAY Lab_Charges
        DISPLAY Medicine_Charges
        DISPLAY Room_Charges
        DISPLAY Subtotal
        DISPLAY Discount
        DISPLAY Final_Bill

    END IF

END
```

9.Source Code

```
# HOSPITAL BILLING AND PATIENT CLASSIFICATION
```

```
def hospital_billing():
    print("\n===== HOSPITAL BILLING SYSTEM =====")

    patient_id = input("Enter Patient ID: ").strip()
    patient_name = input("Enter Patient Name: ").strip()

    if not patient_id or not patient_name:
        print("Error: Patient ID and Name are required.")
        return

    print("\nPatient Categories:")
    print("1. Outpatient (OP)")
    print("2. Inpatient (IP)")
    print("3. Emergency")

    category_choice = input("Select Patient Category (1-3): ").strip()

    categories = {
        "1": "Outpatient (OP)",
        "2": "Inpatient (IP)",
        "3": "Emergency"
    }

    if category_choice not in categories:
        print("Invalid Patient Category.")
        return

    category = categories[category_choice]
```

```
try:
    consultation_fee = float(input("Enter Consultation Fee: "))
    lab_charges = float(input("Enter Laboratory Charges: "))
    medicine_charges = float(input("Enter Medicine Charges: "))
    room_charges = float(input("Enter Room Charges (0 if not applicable): "))
    discount_percent = float(input("Enter Discount Percentage (0-100): "))

    charges = [
        consultation_fee,
        lab_charges,
        medicine_charges,
        room_charges
    ]

    if any(charge < 0 for charge in charges):
        print("Error: Charges cannot be negative.")
        return

    if not 0 <= discount_percent <= 100:
        print("Error: Discount must be between 0 and 100.")
        return

except ValueError:
    print("Error: Please enter valid numeric values.")
    return

subtotal = sum(charges)
```

```
    discount = subtotal * discount_percent / 100
    final_bill = subtotal - discount

    print("\n========== PATIENT BILL ==========")
    print("Patient ID        :", patient_id)
    print("Patient Name      :", patient_name)
    print("Patient Category  :", category)
    print("---------------------------------")
    print("Consultation Fee   : Rs.", format(consultation_fee, ".2f"))
    print("Laboratory Charges  : Rs.", format(lab_charges, ".2f"))
    print("Medicine Charges   : Rs.", format(medicine_charges, ".2f"))
    print("Room Charges       : Rs.", format(room_charges, ".2f"))
    print("---------------------------------")
    print("Subtotal          : Rs.", format(subtotal, ".2f"))
    print("Discount          : Rs.", format(discount, ".2f"))
    print("Final Bill Amount   : Rs.", format(final_bill, ".2f"))
    print("=================================")
    print("Hospital bill generated successfully.")



while True:
    hospital_billing()

    again = input("\nDo you want to enter another patient? (yes/no): ")
    if again.strip().lower() != "yes":
        print("Thank you for using the Hospital Billing System.")
        break
```

10.Test Cases

Test Case ID	Input / Condition	Expected Output	Result
TC01	Valid Patient ID, Name, Category OP, valid charges	Bill generated successfully	Pass
TC02	Valid Patient ID, Name, Category IP, valid charges	Bill generated successfully	Pass
TC03	Valid Patient ID, Name, Category Emergency, valid charges	Bill generated successfully	Pass
TC04	Empty Patient ID	Error: Patient ID and Name are required	Pass
TC05	Empty Patient Name	Error: Patient ID and Name are required	Pass
TC06	Invalid category choice (e.g., 5)	Invalid Patient Category	Pass
TC07	Negative consultation fee or other charge	Error: Charges cannot be negative	Pass
TC08	Non-numeric value entered for charges	Error: Please enter valid numeric values	Pass
TC09	Discount percentage greater than 100	Error: Discount must be between 0 and 100	Pass
TC10	Discount percentage less than 0	Error: Discount must be between 0 and 100	Pass
TC11	All charges are 0 and discount is 0	Final Bill Amount: Rs. 0.00	Pass
TC12	Valid charges with 10% discount	Correct subtotal, discount, and final bill displayed	Pass

RESULT

The test cases verify patient details, patient classification, valid and invalid billing inputs, discount calculation, and final bill generation. The program displays appropriate error messages for invalid inputs and calculates the final bill correctly for valid inputs.

11.Output Screenshots

```
===== HOSPITAL BILLING SYSTEM =====
Enter Patient ID: P101
Enter Patient Name: Ravi

Patient Categories:
1. Outpatient (OP)
2. Inpatient (IP)
3. Emergency
Select Patient Category (1-3): 2
Enter Consultation Fee: 500
Enter Laboratory Charges: 300
Enter Medicine Charges: 700
Enter Room Charges (0 if not applicable): 1000
Enter Discount Percentage (0-100): 10

========== PATIENT BILL ==========
Patient ID          : P101
Patient Name        : Ravi
Patient Category    : Inpatient (IP)
----------------------------------
Consultation Fee    : Rs. 500.00
Laboratory Charges  : Rs. 300.00
Medicine Charges    : Rs. 700.00
Room Charges        : Rs. 1000.00
----------------------------------
Subtotal            : Rs. 2500.00
Discount            : Rs. 250.00
Final Bill Amount   : Rs. 2250.00
==================================
Hospital bill generated successfully.

Do you want to enter another patient? (yes/no): no
Thank you for using the Hospital Billing System.
```

12.Conclusion

The Hospital Billing and Patient Classification system is designed to simplify hospital billing and patient management. It classifies patients into Outpatient, Inpatient, and Emergency categories. The system calculates consultation fees, laboratory charges, medicine charges, room charges, discounts, and the final bill amount. It also validates input data and displays appropriate error messages for invalid entries. This project reduces manual calculations, minimizes billing errors, saves time, and improves the efficiency of hospital administration. Thus, the system provides a simple and effective solution for hospital billing and patient classification.